

A Hybrid Indoor Positioning Solution Based on Wi-Fi, Magnetic Field, and Inertial Navigation

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Abstract—Tracking a pedestrian with a smartphone accurately and in real-time remains a challenging topic in Indoor Positioning Systems. The current computing power of a smartphone is such that it can handle data-intensive applications reasonably well. Additionally, many technologies such as Wi-Fi, Bluetooth, and inertial sensors are already available so no extra hardware cost is needed. However, cost-effective ICs are prone to produce erroneous measurements, resulting in faulty localizations. For instance, Radio Frequency based applications like Wi-Fi and Bluetooth provide accuracy up to 2-3 meters but it requires a lot of effort to maintain the system due to dynamic characteristics of an indoor setting. Additionally, Wi-Fi and Bluetooth ICs in smartphones are usually slow to complete a single scan, which introduces further difficulties when designing real-time tracking applications. Another solution is to use Inertial Measurement Unit (IMU) in a smartphone but accumulative errors due to sensor measurements still remain a problem. In this paper, a hybrid solution is proposed through the exploitation of the unique characteristics of existing technologies and compensating each other's drawbacks. Wi-Fi Fingerprinting is implemented to perform localization when a position of the pedestrian is completely unknown. After narrowing down with the Wi-Fi positioning results, a particle filter, powered by Magnetic Field Fingerprints, is utilized to provide a maintainable and accurate tracking system. Feasibility of the proposed method is demonstrated in an indoor positioning case, where a smartphone device is used throughout the experiment.

Index Terms—Indoor Positioning, Pedestrian Dead Reckoning, Wi-Fi Fingerprinting, Particle Filter, Magnetic Field.

I. INTRODUCTION

Indoor Positioning has become an attractive research field over the last decade and continues to receive a great deal of attention in different scientific disciplines. However, no standard technology has been developed for Indoor Positioning Systems (IPS), unlike, for example, the Global Navigation Satellite System (GNSS) which is commonly used as reliable positioning and navigation technology for outdoor environments. As the number of large complex indoor buildings increases so too does the need for intelligent solutions. Currently, one popular approach is to utilize low-cost and pre-existing technologies. However, one of the reasons why IPS are considered an interesting and complicated problem is the fact that integrat-

ing existing technologies requires interdisciplinary skills and deeper understanding of sensor fusion and signal processing.

This paper addresses the problem of tracking an object in an indoor environment (such as a manufacturing line, an airport or a shopping mall) with smartphones. One possible solution is an anchor based application which exploits Radio Frequency (RF) signals, e.g., Wi-Fi, Bluetooth or radio-frequency identification (RFID) [3] [13]. Most of the time, an indoor environment is covered by plenty of access points or beacons, located in fixed positions, for wireless communication purposes. However, indoor environments tend to be highly dynamic when taking into account human motion, walls or any kind wireless or wired communication networks creating interferences, all of which cause issues, e.g., multi-path propagation, shadowing and reflection [5] [7], in the signal's characteristics. All these issues require re-calibration effort or updating RF fingermaps to provide accurate positioning results. Another solution is dead reckoning systems. Since most of the smartphone consist of Inertial Measurement Unit, movement of the tracking object can be detected. Nonetheless, sensors in IMUs produce noisy and erroneous data, which leads to accumulative errors in the tracking path.

II. PROPOSED METHOD

Many algorithms have been developed for localization purposes throughout the last decade. There are several techniques that can achieve 2-3 m accuracy using different technologies. However, the most challenging task in IPS is still to track a frequently moving object in a dynamic indoor environment, especially if the application targets smartphone users. Therefore, the positioning algorithm must provide accurate results responsively so that users can have a smooth navigation experience. To accomplish such a reliable design, many existing infrastructure technologies must be combined since there is no high accuracy RF positioning system, except costly ultra-wideband radio technologies. Therefore, characteristics of every technology that is easily accessible indoors should be defined in such a way that each technology can be used for a specific purpose based on its unique characteristics.

One of the most widely used IPS technologies is Wi-Fi due to its prevalence in the majority of public and private places. Many researchers have demonstrated that the positioning performance of Wi-Fi suffers from issues like fast fading due to interference if characteristics of radio wave propagation [19] is used to locate mobile devices. While there are fingerprinting methods [2] to avoid such measurement errors, these methods require too much effort in terms of updating fingermaps regularly. However, the bigger concern is the responsiveness because an average Wi-Fi scan duration in a smartphone is approximately 3-5 seconds. This makes it very difficult to track an object in real-time. However, Wi-Fi Fingerprinting technique can be used to locate the object as a region-based approach because it is useful to know in which room or on which floor the tracking object is located within the building.

Another popular approach is to create fingermaps using magnetic field sensors [11]. To distinguish one fingerprint from another, fingerprinting must include clearly distinctive dataset for each fingerprint. In the case of magnetic fields, a particular location might have a very similar distortion somewhere far away from the current location, due to the symmetrical shape of a building composed of very similar ferromagnetic material used in its construction. However, magnetic fields have a unique characteristic when classifying fingerprints within the restricted environment.

In contrast to anchor-based approaches is pedestrian dead reckoning (PDR) [17], which utilizes Inertial Measurement Units (IMU). One does not need to depend on any offline map information to locate itself. However, PDR can only be implemented under very strict conditions. For example, the starting point of the path must be known and all the IMU sensors must provide accurate data. However, IMUs in smartphones are poor at providing long-term accurate results due to the accumulative errors in sensor measurements.

In this paper, the proposed method exploits all of the aforementioned technologies and offers a hybrid IPS solution for tracking. In the proposed method, Wi-Fi Fingerprinting is used for locating the starting point of the PDR algorithm. Subsequently, the particle filter is used to compensate accumulative errors in PDR. Then, Magnetic Field Fingerprinting is used for map constraint by assigning proper weights to particles based on the magnetic field distortion level. With this combination, a reliable IPS algorithm is achieved.

This paper is organized as follows: Section III provides a PDR algorithm for a smartphone application and points out main drawbacks. Section IV shows Wi-Fi Fingerprinting implementation and its classification results. Section V presents a rather unconventional particle filter algorithm using magnetic field as map constraints. Next, Section VI shows test results of the proposed method. Finally, Section VII concludes the paper and notes possible improvements on the proposed method.

III. PEDESTRIAN DEAD RECKONING

The Pedestrian Dead Reckoning (PDR) systems provide a solution for the object tracking problem (1) (2) by attaching an Inertial Measurement Unit (IMU), generally consisting of

accelerometer, gyroscope and magnetic field sensors to a target object. In the past, IMUs were thought to be an external unit which consumes high energy and were mostly used for military or aviation application since it was an expensive hardware. However, this hardware is now available in almost every smartphone and operates at decent energy consumption. To implement a PDR application, the starting point (x_0, y_0) must be known beforehand. Then, tracking of a pedestrian can be done by performing series of step detection algorithms, step length estimations l_t and heading measurements θ_t . It is crucial to detect pedestrian's steps to get an accurate tracking path because any missing step or any drift of orientation will result in an accumulated error.

$$x_t = x_{t-1} + l_t \cos(\theta_t) \quad (1)$$

$$y_t = y_{t-1} + l_t \sin(\theta_t) \quad (2)$$

A. Step Detection

The basic idea of detecting a step taken by a pedestrian holding a smartphone is to measure accelerometer sensor and to observe an increasing and decreasing acceleration pattern [9]. However, the raw acceleration data (3) is too noisy to sufficiently perform any signal processing algorithms. Therefore, pre-processing of the raw acceleration signal is required. First, a high-pass filter (4) is applied to subtract DC component, which is also known as gravity factor. Second, a moving average filter (5) is applied to remove the high-frequency component that can cause misleading information when detecting stride interval.

$$a_{eff,t} = \sqrt{a_{x,t}^2 + a_{y,t}^2 + a_{z,t}^2} \quad (3)$$

$$g_{z,t} = \alpha g_{z,t-1} + (1 - \alpha) a_{eff,t} \quad (4)$$

$$a_{HPPF} = a_{eff,t} - g_{z,t}$$

$$a_{LPPF}[i] = \frac{1}{m} \sum_{j=-(m-1)/2}^{(m-1)/2} a_{HPPF}[i+j] \quad (5)$$

where a_{eff} is raw acceleration of the device to be pre-processed. First, g_z is the gravity factor that subtracted from raw acceleration and α is taken 0.9 while applying the high-pass filter. Next, the low-pass filtered acceleration a_{LPPF} is stored to execute the step detection algorithm and the window size of the moving average filter (m) is taken 10.

Two simple approaches for step detection exist in the literature. The first one is called the zero-crossing method, which assumes that the pedestrian will have positive acceleration until the next step occurs and will count steps whenever acceleration measurement is negative. Based on the collected acceleration data, there are cases when a new step might start before the acceleration crosses zero. Therefore, the second technique, called peak detection [14], is implemented in this paper. The peak detection method assumes that after exceeding the maxima, a measurement below the minima is expected. The goal is to determine stride interval between the last

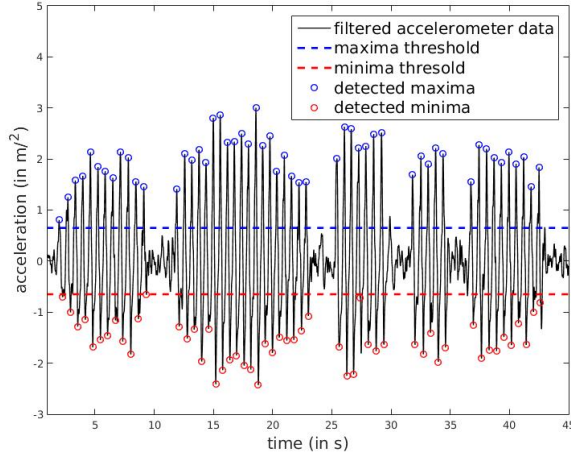


Fig. 1: Peak Detection Technique

updated maxima point and the most recently updated minima point so that the movements of a smartphone that are not a step but perceived as a step can be eliminated. In Figure-1, all the detected maxima values and minima values, one and after, are shown.

For sake of simplicity and performance, maxima and minima thresholds are set empirically. However, it is possible to learn pedestrian's step characteristics and perform classification algorithms to detect steps on different kinds of smartphones or different holding positions of smartphones.

B. Step Length Estimation

There are two main methods to estimate a step length, i.e., static and dynamic. Static methods make an assumption that a pedestrian will have a constant speed, which is not a realistic assumption when the deceleration and acceleration of the movement are considered. In dynamic methods, an acceleration sensor is read during stride interval and is converted into distance using several techniques such as the Weinberg method [18], Kim method [10] and Scarlet method [16]. Weinberg method (6) is chosen due to the fact that it provides the lowest error in comparison to other methods.

$$l = k \sqrt{a_{LPF,peak} - a_{LPF,valley}} \quad (6)$$

where $a_{LPF,peak}$ is the maximum acceleration measured in the Z axis in a single step, whereas $a_{LPF,valley}$ is the minimum acceleration ($k = 0.51$).

C. Heading Estimation

Performing reliable heading estimation indoors [12] has always been a challenging task. One of the most commonly used techniques for estimating orientation is attitude and heading reference systems (AHRS) that fuse magnetic angular rate and gravity sensor arrays (MARG), i.e., accelerometer, gyroscope and magnetic field sensors. Accelerometers and gyroscopes measure physical quantities with respect to the free-fall along one dimension and angular velocity along one rotational axis.

The disadvantage is that header information gathered from acceleration and angular velocity will be drifted over time due to the integration error. On the other hand, magnetic field sensors function as a compass and measure strength of the magnetic field relative to the smartphone's body. The drawback of the magnetic field is that calculated orientation consists of erroneous measurements due to the external magnetic field distortion caused by ferromagnetic materials.

Therefore, one might think that Kalman Filter is the perfect solution to this problem by compensating the integration error (acceleration and gyroscope) with the absolute sensor (magnetic field). This approach works very well outdoors where there are not many ferromagnetic materials. However, many studies [1] and our experiments have shown that the heading angle gathered from magnetic field sensor might provide quite erroneous measurements which can go up to 40 degrees because of the reinforced concrete inside of buildings. These magnetic anomalies clearly worsen the heading drift over time rather than compensating for it. In this paper, the multi-stage compass filter algorithm [8] is used to distinguish good heading measurements from distorted ones. Magnetic field sensor data is rejected for two cases; the one when magnetic field measurement exceeds the reference threshold, and the other when heading from magnetic field sensor does not correlate with the gyroscope's heading. After evaluating magnetic field and deciding that it is a good heading, MARG is fused. In the case of rejection, the heading is calculated from the gyroscope.

IV. WI-FI FINGERPRINTING

RF signals provide attractive solutions for not only communication services but also location-based services. Wi-Fi stands out from rest of other RF technologies that are applied indoors due to ubiquity of Internet access. Many buildings like offices, universities or airports are covered with the Access Points (AP). However, placement of those APs is usually designed for providing sufficient coverage so line-of-sight (LOS) may not be available at some locations. Many researches and experiments are conducted to model propagation characteristics of RF signals indoors. As a result, it is found that many geometric related applications suffer from non-line-of-sight (NLOS). Hence, Wi-Fi Fingerprinting gains an advantage over proximity detection and lateration methods since it only takes into account the uniqueness of the measurements at different locations.

Wi-Fi Fingerprinting requires two phase, i.e., an offline phase and an online phase like in any other fingerprinting methods. In offline phase, $j = 1, 2, \dots, m$ RSSI measurements are collected from $i = 1, 2, \dots, n$ APs on known L_i reference points (RPs) and are stored in the form of multivariate normal distribution $L_i = (\mu_i, \sigma_i)$ into a database:

$$\mu_i = \frac{1}{m} \sum_{j=1}^m x_{i,j} \quad (7)$$

$$\sigma_i = \sqrt{\frac{1}{1-m} \sum_{j=1}^m (x_{i,j} - \mu_i)^2} \quad (8)$$

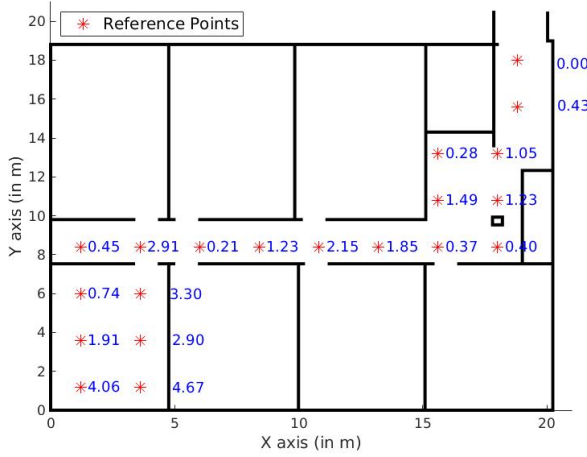


Fig. 2: Wi-Fi Fingerprinting In The Technopark Building

In the online phase, a RSSI measurement is recorded on an unknown location at the time of tracking. Consequently, a RSSI measurement vector s is formed as (9) to perform pattern recognition techniques.

$$s = [AP_1, AP_2, \dots, AP_x] \quad (9)$$

In this paper, the Maximum Likelihood Classification (MLC) algorithm [15] is implemented for Wi-Fi Fingerprinting because RPs consist of independent and unique RSSI values, which maximize the likelihood of the target being in a predicted position, given a set of RPs. To solve this classification problem, comparison between a measurement vector s and RPs stored in database must be done and then one RP having the highest probability $P(L_i|s) > P(L_j|s)$, where $i, j = 1, 2, 3, \dots, n$, $i \neq j$ must be chosen. Each probability is calculated as follows:

$$P(L_i|s) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{s-\mu_i}{\sigma}\right)^2} \quad (10)$$

Wi-Fi Fingerprints are created in a technopark building Figure-2 where 20 RPs, located around 2.4m from each other, are selected. On every RP, 40 Wi-Fi scan results, including APs and RSSIs, are recorded and converted into L_i form. After that, a series of MLC experiments are conducted by performing the classification algorithm and accuracy for each RP is given in meters. Average accuracy of MLC for all RPs on the map is 1.58 m.

This accuracy is obviously not sufficient by itself. In addition, every Wi-Fi scan takes approximately 3-5 seconds depending on the smartphone model. In this case, the pedestrian might already move to another location inside the building within this scan duration. These issues are the biggest drawbacks of the Wi-Fi Fingerprinting method. Nevertheless, another interesting MLC experiment using Wi-Fi is done based on regions. The building is divided into 4 different regions, i.e., Room-1, Hall-1, Hall-2, Exit and then check if MLC algorithm is able to locate the correct region. The test results show that region based Wi-Fi Fingerprinting is able to predict the correct

region 93% of the time. Therefore, initialization of the PDR, where the position is completely unknown, is calculated by region based Wi-Fi Fingerprinting.

V. PARTICLE FILTER

Particle filter is a numerical approximation to the recursive Bayesian Filter, which is also called to Sequential Monte Carlo Estimation. Unlike the Kalman Filter, a particle filter only considers posterior state $S_t = (s_t^i, w_t^i)$, $i = 1, 2, 3, \dots, n$ where s_t^i represents state variables and w_t^i stands for weight. This makes a particle filter an effective solution for tracking because the biggest issue of the tracking scenarios is the accumulated heading error caused by non-Gaussian-distributed noise. PDR performs quite sufficiently for a short period of time but it suffers from heading drift and step length estimation error. In this case, a particle filter can be utilized as a recovery mechanism when unintended tracking paths occur. Whenever a new step is detected, the previous posterior state is updated with the current state information, each particle is weighted based on the new information, and the current particle set is resampled to ensure that they are distributed on their likely positions.

For particle filter [4] to be an accurate positioning algorithm, map constraints must be included as a state variable so that particles can be weighted accordingly. Mostly, the building plan is known and particles' weight is assigned low whenever they are updated in a way that particles that are less likely to be real location begin to violate rules like crossing wall behaviors so that they will be filtered out throughout the process of resampling. In this paper, a different type of map constraint is introduced by using magnetic field information:

$$s_t^i = (x_t, y_t, \theta_t, B_{x,t}^{GCS}, B_{y,t}^{GCS}, B_{z,t}^{GCS}) \quad (11)$$

where x_t, y_t are location coordinates in the building, θ_t is heading angle, and $B_{x,t}^{GCS}, B_{y,t}^{GCS}, B_{z,t}^{GCS}$ are magnetic field information that is converted from local coordinate system (LCS) to global coordinate system (GCS) [9]. Each of these state variables represents a particle.

However, the challenging part is to assign magnetic field for particles since there is no way of making adequate assumptions or calculations without obtaining the indoor characteristic of the magnetic field. Thus, a high-resolution magnetic field fingerprint, shown in Figure-3, is created by collecting data at every 40 cm in the technopark building. Magnetic field sensor is scanned 40 times for each RP while holding the phone in the hand. Concurrently, the device orientation is also gathered by using the multi-stage compass filter algorithm [8] and the magnetic field is converted from LCS to GCS. Then, the magnetic field fingerprint is saved into the database.

Assigning initial coordinates (x_0, y_0) to each particle, so-called particle spreading, is executed randomly. In our case, particle spreading is limited to the region based Wi-Fi Fingerprinting result. Before assigning a random location to a particle, Wi-Fi Fingerprinting is performed. Then, particles are created randomly inside the corresponding region so that the possibility of a particle to be the real location is increased. On the other hand, initial heading (θ_0) for each particle is assigned

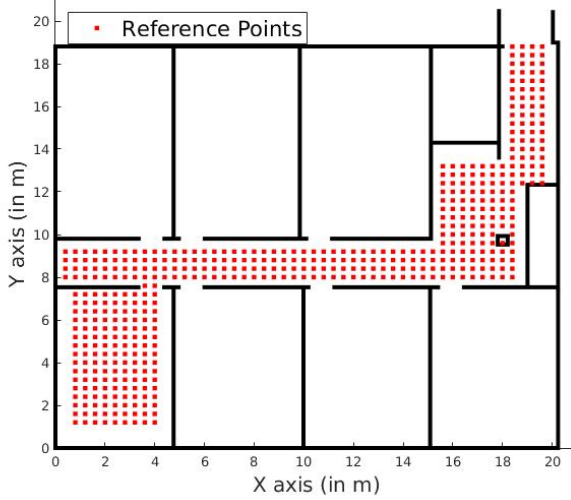


Fig. 3: Magnetic Field Fingerprintmap In The Technopark Building

based on the azimuth angle that is read from the smartphone's magnetic field sensor along with added random noise. After the initialization procedure is completed, the coordinates and heading of particles are updated with the PDR algorithm.

After coordinates and heading for each particle are initialized, the particles will hold possible locations on the map. With this information, a correlation between a particle and an RP can be defined. The initial magnetic field information ($B_{x,0}^{GCS}$, $B_{y,0}^{GCS}$, $B_{z,0}^{GCS}$) of the particles are assigned based on the adjacent RP's magnetic field value that is collected as a magnetic field fingerprintmap. Each particle is a neighbor to a group of RPs and magnetic field value for the particle is taken from the RP that has the shortest Euclidean distance to the particle. On every update on particles' coordinates, assigning magnetic field information to the particles is executed in the same way. As a result, a particle filter can be structured as 3 fundamental steps, i.e., update, weight and resample.

In update step, step length l_k and heading angle difference $\Delta\theta_k$ between a current state and previous state are added to particles' previous state coordinates. n_0 and n_1 represent measurement errors of heading angle and step length respectively. Additionally, these errors are added during update step as zero-mean Gaussian random noises.

$$x_t = x_{t-1} + (l_t + n_1)\cos(\Delta\theta_t + n_0) \quad (12)$$

$$y_t = y_{t-1} + (l_t + n_1)\sin(\Delta\theta_t + n_0) \quad (13)$$

$$\theta_t = \theta_{t-1} + \Delta\theta_t + n_0 \quad (14)$$

In weight step, the main idea is to assign a weight to each particle based on magnetic field values. While PDR algorithm operates to detect steps, magnetic field measurements are also collected concurrently. The weighting process is done with the help of Maximum Likelihood Normal Distribution (10) so that particles that diverge from measured magnetic field value during walking will be filtered out over the course of pedestrian's movement. The only difference is that the logarithmic result of the Maximum Likelihood is normalized

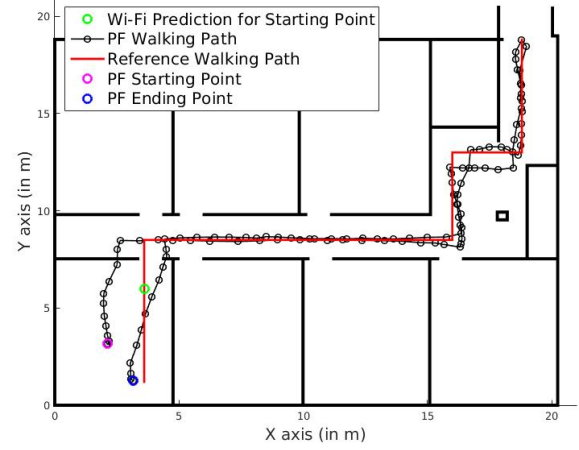


Fig. 4: Test Results of Proposed Particle Filter

to 1 and is assigned as a weight. As a result, meaningful comparison between measured magnetic field and particles' magnetic field is achieved.

In resample step, updated and weighted particles that obtain the low probability of the real location must be eliminated so that maximum likelihood particles can gather around the pedestrian's position. Many sophisticated approaches exist in literature but Multinomial Resampling [6], which is a simple random selection algorithm, is chosen due to performance issues in smartphones. Note that resampling process may cause uncertainties because particles might be distributed unevenly on the building over time. Hence, Multinomial method select N particles randomly to check if the weight is under the desired threshold.

After these 3 steps are concluded, current state's position is calculated by taking an average of all particles. This process continues iteratively whenever PDR detects a step.

VI. TEST RESULTS

The algorithm that is proposed in this paper is tested with Sony Xperia Z2 smartphone. A reference path, which has 65.6 meters length, is walked with the phone facing toward the pedestrian. The reference path starts at $(X, Y) = (3.6m, 1.2m)$, reaches up til $(X, Y) = (18.8m, 18m)$ and then returns back to the starting point. Whenever the PDR algorithm detects a step, magnetic field data are stored in the phone to perform a particle filter algorithm. The total number of steps taken 107 and PDR's step detection algorithm counts as 108. Since starting point is decided by Wi-Fi Fingerprinting, there is a possibility that starting point might be set erroneously up to 5m meters sometimes.

In Figure-4, one of the worst-case scenarios is shown to point out how proposed particle filter is a powerful recovery technique to track moving objects indoors. $(X, Y) = (3.6m, 1.2m)$ is the real starting point of the walking path but Wi-Fi Classification is calculated as $(X, Y) = (3.6m, 6m)$ on the map. Then, particle filter's tracking path is started at $(X, Y) = (2.12m, 3.17m)$ after creating particles on only

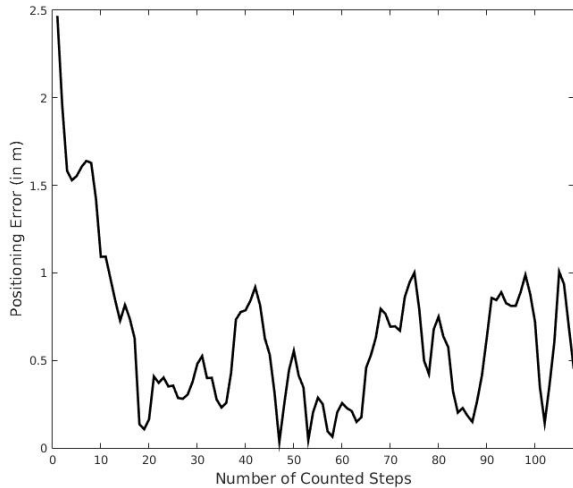


Fig. 5: Positioning Error of Proposed Particle Filter

the predicted room and weighting particles based on the measured magnetic field. Therefore, prediction of starting point using Wi-Fi Fingerprinting is already improved by 2.88 m from the start. Finally, the walking path is ended at $(X, Y) = (3.15m, 1.26m)$. Total error difference between the reference path and the predicted path is shown in Figure-5. As a consequence, although Wi-Fi Fingerprinting predicts the starting point with a 4.8m error, the proposed particle filter manages to keep the error under 1m after 10 steps and to end with a 0.45m error.

VII. CONCLUSION

This paper proposes a hybrid solution to tracking with smartphones indoors. After existing technologies for IPS have been studied comprehensively, drawbacks and strengths of these technologies have been defined. It is demonstrated that average localization error of Wi-Fi is 1.58 m and Wi-Fi suffers from long scan duration due to cost-effective ICs in smartphones. Nevertheless, Wi-Fi is good at finding region-based locations so it can serve as a starting point provider for PDR and particle filters when the location is completely unknown. As discussed in Section III, PDR has a drifting effect that requires an additional technique to compensate for the error. Therefore, the proposed particle filter, weighting each particle based on magnetic field information, is implemented. Inevitably, all of the sensor fusion methods mentioned in this paper require high computation power, which is quite limited in smartphones for particle filter techniques. In future work, particle filters, especially including high number particles, must be optimized to perform without any lags caused by computation delays since the desired navigation application should operate responsively and seamlessly.

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